

Yixing Wu

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Summary

Ph.D. student in Electrical Engineering at USC, working on **Generative AI and large-scale language model systems**. Hands-on experience with **Transformer-based models**, **distributed training**, and **efficient inference** for long-context generation. Interested in **Machine Learning Engineer / Applied Scientist Intern** roles focused on scalable ML and AI systems.

Research Interests

Generative AI & LLM Systems: Training and inference optimization for Transformer-based models, including distributed training, long-context modeling, and efficient decoding.

Efficient and Scalable Machine Learning: Resource-aware learning pipelines for high-dimensional data, long sequences, and large-scale inference systems.

Education

University of Southern California Ph.D. Student in Electrical Engineering, Minor area in Computer Science Advisor: Professor C.-C. Jay Kuo	Los Angeles, United States 08/2024 - Present
University of Southern California M.S. in Electrical Engineering (Honors)	Los Angeles, United States 01/2022 - 12/2023
Xidian University B.S. in Information Engineering	Xi'an, China 09/2016 - 07/2020

Selected Publications (Full list at: [\[Google Scholar\]](#))

- Dingyi Nie, **Yixing Wu**, C.-C. Jay Kuo. *Modeling Irregular Multivariate Time Series with Missing Observations*. Accepted at APSIPA Transactions on Signal and Information Processing. Robust learning for long time-series under irregular sampling and missing data.
- Xinyu Wang, **Yixing Wu**, et al. *A Statistics-based Feature Generation (SFG) Method: Theory and Applications*. 2024 IEEE International Conference on Big Data (BigData), 5731–5738. Efficient feature generation and selection for high-dimensional learning systems.
- Wei Wang, **Yixing Wu**, C.-C. Jay Kuo. *A Green Learning Approach to LDCT Image Restoration*. 2025 IEEE International Conference on Image Processing (ICIP), 1762–1767. Efficient image generative modeling for inverse problems with reduced computation and inference cost.

Awards

USC Viterbi Research Assistantship	08/2024 - Present
USC EE MS honors program	09/2023 - 12/2023

Selected Research Projects

Large-Scale Language Model Training and Inference	01/2025 – Present
- Built decoder-only Transformer models with multi-GPU training using PyTorch Distributed Data Parallel. - Implemented KV-cache and Rotary Positional Embeddings for efficient long-context autoregressive decoding. - Analyzed accuracy–latency–memory trade-offs under different decoding strategies.	
Hierarchical Models for Long Time-Series Event Detection	06/2025 – 09/2025
- Built multi-resolution models for event detection in long and noisy time-series. - Applied coarse-to-fine inference to capture long-range dependencies efficiently. - Improved training stability and scalability by avoiding backpropagation through long sequences.	

Generative Modeling for Image Restoration

01/2025 – 03/2025

- Developed generative models for robust image reconstruction under severe noise.
- Designed multi-stage, coarse-to-fine pipelines with improved stability and efficiency.
- Achieved competitive performance compared to deep learning baselines with smaller models.

Technologies

- **Programming:** Python, C/C++ , Java, MATLAB
- **ML Frameworks:** PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost
- **Systems & Tools:** Git, Docker, Linux